

# Investigating with commands

NUnix · L3 MIAGE · CM2 · 90 minutes

ISTIC · University of Rennes 2026-2027



## Investigating with commands



A computer is slow. Would adding RAM help?  
Check which resource is under pressure first.

- 1. Check CPU, memory and disk usage.
- 2. Find large files and inspect application logs.
- 3. Count error codes with a pipeline.

## 1. Processes and resources



Check CPU, memory and disk usage

## CPU, memory and disk space



| Resource | Question                                 | First observation |
|----------|------------------------------------------|-------------------|
| CPU      | Is computation keeping a processor busy? | top, ps           |
| RAM      | Is memory pressure affecting the system? | free -h           |
| Disk     | Can the filesystem store more data?      | df -h .           |

Adding RAM does not create free space on a full disk.

## Running in the foreground



A **process** is a running instance of a program.

```
sleep 10
```

The shell waits for this foreground command to finish before showing a new prompt.

**Ctrl+C** requests an interrupt of the foreground job.

## Inspecting processes



```
ps
ps -ef
```

| Field | Meaning                     |
|-------|-----------------------------|
| PID   | Process identifier          |
| TTY   | Associated terminal, if any |
| TIME  | Accumulated CPU time        |
| CMD   | Command                     |

### Checking memory usage



```
free -h
ps -eo pid,comm,%cpu,%mem --sort=-%mem
```

- ▶ Read **available** memory; cache use is normal.
- ▶ Identify the application's process and follow it over time.
- ▶ Check logs before attributing every delay to memory.

7/42

### Checking disk usage



| Tool | Scope              | Question                     |
|------|--------------------|------------------------------|
| df   | A filesystem       | How much space is available? |
| du   | A directory tree   | Which directories use space? |
| find | Individual entries | Which files fit my criteria? |

Start broad, then inspect a **specific file**.

8/42

### Find which directories use space



```
df -h "$HOME"
du -h --max-depth=1 "$HOME"
```

df shows usage of the filesystem containing your home.  
du shows how much space your directories use.

9/42

### Finding files



```
find . -type f -name '*.txt'
```

- ▶ **.** is the **starting directory**.
- ▶ **find** also visits its subdirectories.
- ▶ **-type f** selects regular files.
- ▶ **-name '\*.txt'** tests each entry's name.

The quotes let **find** receive the pattern unchanged.

10/42

### Search criteria



| Criterion       | Matches                         |
|-----------------|---------------------------------|
| -iname '*.txt'  | Name, ignoring case             |
| -user alice     | Entries owned by Alice          |
| -type d/-type l | Directories / symbolic links    |
| -size +100M     | Size greater than 100 MiB       |
| -mtime -2       | Modified less than 48 hours ago |

```
find . -type f -size +100M
```

11/42

### Checkpoint: find



1. Find regular **.c** files below the current directory.
  2. Why quote **'\*.c'**?
1. `find . -type f -name '*.c'`
  2. So the shell passes the pattern to find without expanding it first.

12/42

## 2. Searching text



Read log records and search for errors

13/42

## A small log file



Save these four lines as `app.log`:

```
10:00 INFO STARTED
10:01 ERROR DISK_FULL
10:02 ERROR TIMEOUT
10:03 ERROR DISK_FULL
```

| Field | Meaning    |
|-------|------------|
| 1     | Time       |
| 2     | Severity   |
| 3     | Event code |

14/42

## Finding text with grep



```
grep 'ERROR' app.log
grep -r 'ERROR' .
grep -n 'DISK_FULL' app.log
```

- ▶ Print **matching lines** from a file.
- ▶ `-r` searches directories recursively.
- ▶ `-n` includes line numbers.

Exit status: **0** = match, **1** = no match, **2** = error in GNU grep.

15/42

## Where the name grep comes from



The editor `ed` had the command pattern:

```
g/re/p
```

- ▶ **g**: apply globally to matching lines.
- ▶ **re**: a regular expression.
- ▶ **p**: print those lines.

`grep` makes that text search available as a separate command.

16/42

## Patterns in grep



| Regular expression | Meaning                                    |
|--------------------|--------------------------------------------|
| <code>.</code>     | Any one character                          |
| <code>m*</code>    | Zero or more repetitions of <code>m</code> |
| <code>[1-3]</code> | A digit from 1 to 3                        |

```
grep 'm.*t' names.txt
grep 'toto[1-3]' names.txt
```

In a regular expression, `*` repeats the **preceding item**.

17/42

## Who interprets this pattern?



```
printf '%s\n' *.log
grep 'DISK.*' app.log
find . -name '*.log'
```

1. The **shell** expands a filename pattern.
2. **grep** interprets a regular expression.
3. **find** interprets a quoted filename pattern.

18/42

## Input, output and status



| Channel | Number   | Usual terminal behaviour    |
|---------|----------|-----------------------------|
| stdin   | <b>0</b> | Read keyboard input         |
| stdout  | <b>1</b> | Display normal output       |
| stderr  | <b>2</b> | Display diagnostic messages |

The **exit status** is a number reported when a command finishes. It is separate from the text it prints.

19/42

## Exit status



```
true
echo "$?"
false
echo "$?"
```

**0** means success / true. **Nonzero** means failure / false.  
\$? is the status of the **most recent command**.

20/42

## 3. Redirections and pipes



Choose where a command reads and writes

21/42

## Reading input from a file



```
sort < names.txt
```

The shell opens names.txt as **standard input** for sort.

sort writes the sorted lines to standard output, which still points to the terminal.

22/42

## Saving output



```
sort names.txt > sorted.txt
echo 'Zoé' >> sorted.txt
```

| Operator | Effect on the destination file              |
|----------|---------------------------------------------|
| >        | Create it, or <b>replace</b> its contents   |
| >>       | Create it, or <b>append</b> to its contents |

23/42

## Redirecting errors



```
find . -name '*.txt' > matches.txt 2> errors.txt
```

- ▶ > redirects **stdout**, descriptor 1.
- ▶ 2> redirects **stderr**, descriptor 2.
- ▶ Normal results and diagnostics go to separate files.

24/42

## Redirections are applied from left to right



```
ls notes.txt missing > all.txt 2>&1
ls notes.txt missing 2>&1 > out.txt
```

| Command | stdout ends up... | stderr ends up... |
|---------|-------------------|-------------------|
| First   | In all.txt        | In all.txt        |
| Second  | In out.txt        | At the terminal   |

2>&1 copies stdout's destination **at that moment**.  
It does not make stderr follow later changes to stdout.

25/42

## Why did the file become empty?



```
sort names.txt > names.txt
```

The shell opens and truncates names.txt **before** starting sort. The input is already gone.

Write to a **different file**, then inspect the result.

26/42

## A pipe between two commands



```
grep 'ERROR' app.log | wc -l
```

grep writes matching lines to its standard output.  
The pipe connects that output to wc's standard input.  
The result is the **number of matching lines**.

27/42

## Connecting programs like garden hoses



Doug McIlroy described connecting programs so that one program's output could become another's input.

```
grep 'ERROR' app.log | wc -l
```

Each tool has a small job.  
The shared text stream lets us combine them.

28/42

## 1. Select errors and extract their codes



```
grep 'ERROR' app.log |
cut -d ' ' -f 3
```

```
DISK_FULL
TIMEOUT
DISK_FULL
```

29/42

## 2. Count each error code



```
grep 'ERROR' app.log |
cut -d ' ' -f 3 |
sort | uniq -c | sort -nr
```

```
2 DISK_FULL
1 TIMEOUT
```

uniq -c counts **adjacent** equal lines. Sort first.

30/42

## Save and display output with tee



```
sort names.txt | tee sorted.txt
```

tee copies its input to a file **and** standard output.

Think of a T-shaped pipe fitting: the stream has two outlets.

31/42

## Combining arguments and redirections



```
sort < names.txt > sorted.txt
sort < names.txt | head -n 3 > first-three.txt
```

- ▶ < names.txt selects the input stream.
- ▶ -n 3 remains an **argument** to head.
- ▶ > first-three.txt selects the output file.

32/42

## Checkpoint: redirections



- |                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| 1. Which operator appends to a file?              | 1. >>                             |
| 2. What does a   b connect?                       | 2. a's stdout to b's stdin.       |
| 3. Does a > out.txt also redirect standard error? | 3. No. Use 2> to redirect stderr. |

33/42

## The exit status of a pipeline



```
false | sort
echo "$?"
set -o pipefail
false | sort
echo "$?"
```

Output: **0**, then **1**.  
Bash normally reports the last pipeline command's status.

34/42

## CM2 summary: commands and pipelines



- ▶ Check **CPU**, **memory** and **disk** usage.
- ▶ Find files and filter lines with **grep**.
- ▶ Combine commands with pipes and redirections.

35/42

## Optional reference



More search expressions and interactive process control

36/42

### Combining criteria



```
find /usr -name '*.c'
find . -user alice -name '*.c'
find . -not \( -name '*.obj' -o -name '*.map' \)
```

- ▶ Adjacent tests use **AND**. -a makes it explicit.
- ▶ -o means **OR**. -not negates a test.
- ▶ Escaped parentheses group tests for find.

37/42

### Running a command on matches



```
find . -type f -name '*.txt' -exec wc -l {} \;
```

- ▶ -exec starts the command to run.
- ▶ {} stands for the current matching path.
- ▶ \; ends the command without ending the shell's command line.

38/42

### Running in the background



```
sleep 60 &
jobs
fg %1
```

- ▶ & lets the shell return to the prompt immediately.
- ▶ jobs lists jobs managed by this shell.
- ▶ fg %1 brings job 1 to the foreground.

39/42

### Suspending and resuming a job



| Action        | Effect                         |
|---------------|--------------------------------|
| <b>Ctrl+Z</b> | Suspend the foreground job     |
| bg %1         | Resume job 1 in the background |
| fg %1         | Resume job 1 in the foreground |

A **job number** belongs to a shell.  
A **PID** identifies a process in the system.

40/42

### Sequential and asynchronous execution



```
who | sort; date
sleep 10 &
date
```

- ▶ ; runs the next command after the previous one finishes.
- ▶ & starts work asynchronously.
- ▶ The final date can run while sleep is still active.

41/42

### Signals and end of input



| Action        | Meaning                                      |
|---------------|----------------------------------------------|
| kill PID      | Send SIGTERM to a process                    |
| <b>Ctrl+C</b> | Send SIGINT to the foreground process group  |
| Ctrl+\        | Send SIGQUIT to the foreground process group |
| <b>Ctrl+Z</b> | Send SIGTSTP to suspend the foreground group |
| <b>Ctrl+D</b> | Terminal end-of-input action                 |

42/42